

Ammonia-Fired Power Generation — OEM Survey

MYSGH2 Project (Malaysia → Singapore Green Hydrogen via Green Ammonia) | Gas turbines, 2-stroke & 4-stroke gas engines | Last verified 2026-09-10

OEM	Gas Turbine (ammonia-fired solution / status)	2-Stroke Gas Engine (ammonia-fired solution / status)	4-Stroke Gas Engine (ammonia-fired solution / status)	News link
GE Vernova	F-class 100 % NH3 combustion system (with IHI). Full-scale F-class combustor components run on 100 % ammonia at full-load conditions at IHI's Large-Scale Combustion Test facility, Aioi, Hyogo (facility inaugurated Jun 2025); announced Mar 2026 . Roadmap: commercial ammonia-capable F-class turbine by 2030 , as a retrofittable combustion system for the installed 6F.03 / 7F / 9F fleet. JDA signed 2024; MoU Jun 2021. Prototype-combustor testing continuing. Emissions "align with the joint roadmap" -- no NOx/N2O numbers published . *Status: demonstrated at full-scale component level; pre-commercial.*	No known program	No known program	https://www.gevernova.com/news/press-releases/ihi-ge-vernova-achieve-milestone-100-ammonia https://www.gevernova.com/gas-power/future-of-energy/ammonia-powered-gas-turbines
IHI Corporation (incl. IHI Power Systems, ex-Niigata)	IM270, 2 MW class -- the most operationally proven 100 %-NH3 machine on either technology . World-first 100 % liquid ammonia combustion Jun 2022; package demo (2 MWe + heat-recovery boiler) at Aioi from Jul 2024, ~3,000 h as of Jan 2026 , incl. load-rejection/load-dump. NOx <7 ppm (after DeNOx); >99 % GHG cut incl. N2O at 70-100 % NH3 via two-stage rich-lean combustion. AEA reports IHI targeting commercialisation 2026. First commercial-plant deployment: 2 MW IM270 at PETRONAS' integrated chemical complex, Terengganu, Malaysia -- JCDA with PETRONAS + Gentari , demo start 2027 . Also the F-class partner to GE Vernova (left column).	No known program	Land-based ammonia genset -- 6,000 kW class `18V28ADF` . Demonstration plant commenced operation at IHI Power Systems' Ota Works, Gunma (announced 2026 FY): ammonia/heavy-fuel-oil dual fuel, also 100 %-HFO capable; target >90 % ammonia fuel ratio and >90 % GHG reduction ; demo completion JFY2026, commercial sales from JFY2027 ; explicitly aimed at data centres, industrial parks, remote islands, mining . Marine ancestor: 4-stroke ammonia DF engine on tug *Sakigake* in service Aug 2024; CIMAC 2025 President's Award. *This is the only stationary ammonia genset with an announced commercial sales date.*	https://www.ihi.co.jp/en/all_news/2026/resources_energy_environment/1202137_13814.html https://ammoniaenergy.org/articles/ihi-petronas-to-demonstrate-ammonia-fired-gas-turbine-in-malaysia/
Mitsubishi Heavy Industries / Mitsubishi Power	H-25, 40 MW class, 100 % ammonia direct firing -- development announced 1 Mar 2021 , "commercialisation in or around 2025 "; approach = newly developed low-NOx combustor + SCR. No MHI release located (2023-2026) confirming, revising or retracting that date -> treat as stale/unconfirmed , not as a confirmed slip. Demonstrated hardware to date is 20 % NH3 (LHV) co-firing on a 2 MW-class machine (Yokohama, Oct 2020), raw NOx 287 ppm @16 % O2 -> <7 ppm only with a NOx-removal device. Joint R&D with Institut Teknologi Bandung (Indonesia) on ammonia-fired GT power, signed Sep 2022, extended Nov 2023. Separate track (not a GT): ammonia single-fuel boiler burner , 0.5 t/h, Nagasaki, Nov 2023.	No known program	No known program	https://power.mhi.com/news/20210301.html https://www.powermag.com/mitsubishi-power-developing-100-ammonia-capable-gas-turbine/
Kawasaki Heavy Industries	No known ammonia-fired gas turbine program . KHI's decarbonisation turbine effort is hydrogen : 100 % H2 DLE combustion, GPB17MMX 1.8 MW-class H2 turbine (Hydrogen Technology of the Year 2024), RWE Lingen 100 %-H2-capable project. *No ammonia GT milestone located as of Sep 2026 -- flagged as "not found", not as "does not exist".*	No known program	No known ammonia program . Kawasaki Green Gas Engine track is hydrogen: world-first commercial launch of a 30 % H2 co-firing large gas engine, 30 Sep 2025 .	https://global.kawasaki.com/en/corp/profile/division/gasturbine/news.html https://www.kawasaki-energysolution.com/gasengine_gasturbine/en/index.html
Siemens Energy	No ammonia-fired product or named machine demonstration located . Ammonia is listed among "sustainable fuels" for gas turbines alongside H2/methanol/biogenic fuels, and Siemens Energy participates in academic work on partially decomposed (cracked) ammonia in scaled burners; separately it backs ammonia cracking ventures (NH3->H2 upstream of an H2 turbine). No SGT model, blend %, or demo date published . *Status: research / fuel-flexibility positioning only.*	No known program	No known program	https://www.siemens-energy.com/global/en/home/products-services/solutions-usecase/sustainable-fuels.html https://www.siemens-energy.com/us/en/home/innovations/ventures/ammonia-cracking.html
Ansaldo Energia	No ammonia-fired hardware program located . Ansaldo's carbon-free turbine effort is 100 % hydrogen on the GT36 constant-pressure sequential combustor (FLEX4H2; full-scale can tests at DLR Cologne / HELENA rig). Ammonia appears only as a numerical concept study -- part-load operation of the sequential combustion system in rich-dilute-lean mode on ammonia-based gaseous fuels (ASME *J. Eng. Gas Turbines Power*, vol. 148). *Status: simulation / paper stage.*	No known program	No known program	https://asmedigitalcollection.asme.org/gasturbinespower/article-abstract/148/5/051015/1222396/Numerical-Concept-Study-for-Part-Load-Operation-of https://www.ansaldoenergia.com/about-us/media-center/power-generation-news-insights/detail-news/gt36-sequential-combustion-technology-achieves-100-hydrogen
Baker Hughes	NovoLT16 (~16 MW class), 100 %-ammonia-capable dual-fuel turbine -- JDCA with Hanwha Power Systems + Hanwha Ocean , announced Feb 2025 (Baker Hughes Annual Meeting, Florence). Baker Hughes supplies the turbine; Hanwha the ammonia combustion system (PoC combustor already tested on 100 % NH3). Baker Hughes turbine feasibility studies completed 2024. Full engine test on 100 % ammonia targeted by end-2027 , commercial orders thereafter; deployment 2028. Marine and onshore/offshore, power generation and mechanical drive. Vendor goal: efficiency "comparable to reciprocating engines"; claims IMO Tier III NOx without SCR -- *vendor claim, unverified at stationary duty.*	No known program	No known program	https://investors.bakerhughes.com/news-releases/news-release-details/baker-hughes-and-hanwha-announce-partnership-development-small-size https://gasturbineworld.com/hanwha-combustion-systems/
Hanwha (Power Systems / Ocean / Engine)	Ammonia combustion system for the Baker Hughes NovoLT16 (see row above) -- Hanwha owns the combustor, Baker Hughes the turbomachinery. 100 %-NH3 combustor proof-of-concept already tested . Hanwha Group targets ammonia-turbine-propelled LNG carriers and containerships by 2028 .	Unconfirmed -- Hanwha Engine (ex-HSD) is a licensed two-stroke builder, but no dedicated Hanwha ammonia two-stroke program was sourced in this survey. Do not assume.	No known program	https://www.hanwha.com/newsroom/news/press-releases/hanwha-and-baker-hughes-announce-agreement-for-joint-development-of-ammonia-gas-turbines-do https://lngprime.com/asia/hanwha-baker-hughes-to-develop-ammonia-turbines/139641/

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Doosan Enerbility	No direct ammonia-fired turbine. Doosan's ammonia pathway is crack first, fire hydrogen : a 380 MW H-class 100 %-hydrogen turbine (target ~2027) fed by ammonia cracking -- Johnson Matthey supplying cracking technology/catalysts; separate ammonia-cracked-hydrogen turbine/combustor work with POSCO, RIST and KEPCO . Separate (boiler, not GT) track: ammonia co-firing burner , Indonesia Power Suralaya study, burner commercialisation targeted 2027 .	No known program	No known program	https://www.doosanenerbility.com/en/about/news_board_view?id=21000333 https://gasturbineworld.com/doosan-gas-turbines/
Solar Turbines (Caterpillar)	No known ammonia program. No Solar Turbines ammonia announcement, demonstration or product page located as of Sep 2026. (Generic ammonia-GT patents exist in the public record but were not attributable to Solar in this search -- excluded.)	No known program	Caterpillar Energy Solutions / MWM: no known ammonia program -- gas-engine roadmap is natural gas / biogas / hydrogen.	https://www.solarturbines.com/en_US/about-us.html
Everllence (formerly MAN Energy Solutions)	No known program	B&W `ME-LGIA` -- the reference commercial ammonia two-stroke. Diesel-cycle liquid ammonia injection, ~5 % pilot ("SPOC"), >95 % ammonia energy fraction. Full-scale testing on the 4T50ME-X test engine, Copenhagen; Type Approval Test passed Jun 2026; first engine delivery Q1 2026 ; market introduction of 50-, 60-, 70- and 80-bore variants expected by end-2026 . Pilot projects: 2× VLACs (Eastern Pacific Shipping), 4× Höegh Autoliners PCTCs, 1× Japanese bulker. MAN two-stroke portfolio spans 12-68 MW . *Marine-first; no stationary variant announced.*	`NH3 Spark - FutureFlex` -- dual-fuel four-stroke GenSet running on ammonia with NO pilot fuel (a first for a commercial industrial engine), led by MAN ES with DTU and Skovgaard Energy . Deliberately small-bore , prioritising simplicity/price/ retrofit into the >20,000 MAN GenSets in service. Phase 1: single-cylinder ammonia testing at DTU Construct from Q3 2025 ; Phase 2: full-scale testing under power-plant conditions at Skovgaard's green-ammonia facility . Predecessor programme *AmmoniaMot 2* (Aug 2024, ~3.5 yr). *Status: R&D; no product date.*	https://www.everllence.com/company/press-releases/details/2026/04/20/ammonia-engine-passes--huge-milestone--en-route-to-commercial-debut https://www.man-es.com/company/press-releases/press-details/2025/05/22/man-energy-solutions-to-lead--new-ammonia-genset-project
WinGD	No known program	`X-DF-A` -- first ammonia two-stroke in commercial service. X52DF-A: Type Approval Test + FAT completed Jan 2026; world's first ocean-going ammonia-fuelled vessel, *ANTWERPEN* , delivered to EXMAR Jun 2026 (built HD Hyundai Mipo; engine built by HD Hyundai Heavy Industries Engine & Machinery). ~30 X-DF-A engines on order across gas carriers, bulkers, tankers, container vessels. *Marine only; no stationary offering.*	No known program	https://wingd.com/news-media/news/wingd-powers-delivery-of-world-s-first-ocean-going-ammonia-fuelled-vessel-antwerpen-to-ex-mar https://wingd.com/news-media/news/wingd-achieves-first-type-approval-and-factory-acceptance-testing-for-ammonia-fuelled-two-stroke-engine
Wärtsilä	No known program	No known program	`Wärtsilä 25 Ammonia`, 1.9-3.1 MW/unit . Type-approval tests under class supervision autumn 2025 ; power upgrade announced 16 Apr 2026 (315 kW/cyl @900 rpm, 345 kW/cyl @1000 rpm) . Up to 90 % GHG reduction at ~95 % ammonia energy share (~5 % pilot). Package includes AmmoniaPac fuel supply, WARMS release-mitigation, NOR (SCR) NOx reducer. [!] Delivery year conflict -- unresolved : repo memory records deliveries 2028 ; a Sep-2026 reading of the same Wärtsilä release states "inaugural deliveries due from 2026 onward." Do not quote either without re-checking the primary release. For stationary plants Wärtsilä today offers hydrogen conversion packages and states only a *capability* to develop ammonia conversions "when required" -- i.e. no stationary ammonia product ; data-centre offering remains natural gas (34SG/50SG).	https://www.wartsila.com/usa/media/news/16-04-2026-power-increase-for-wartsila-25-ammonia-engine-supports-more-efficient-ammonia-fuelled-vessel-designs-3739966 https://www.wartsila.com/energy/sustainable-fuels/future-proofing-flexible-wartsila-engine-power-plants-for-decarbonised-energy-systems
HD Hyundai (HHI Engine & Machinery / HD Hyundai Heavy)	No known program	Licensed builder, not developer : HD Hyundai Heavy Industries Engine & Machinery built the WinGD X52DF-A ammonia two-stroke installed on *ANTWERPEN* (world-first ammonia-fuelled ocean-going ship, Jun 2026). No in-house HD Hyundai ammonia two-stroke design sourced.	HiMSEN ammonia dual-fuel, high-pressure direct injection (world-first) . `H22CDF-LA`: model-based class approval completed Oct 2024 , commercialisation begun , output to 2.2 MW ; approvals from 7 major classification societies . `H32CDF-LA` (Oct 2025): same high-pressure DI technology on the H32C platform, output to 5.4 MW . Claimed ~90 % GHG reduction vs. diesel. HD Hyundai states applications include marine gensets/propulsion and plans to supply the land-based power-generation market .	https://www.hyundai-engine.com/en/media/newsdetail/435 https://fuelcellsworks.com/2025/10/20/energy-innovation/hd-hyundai-heavy-industries-engine-and-machinery-develops-4-stroke-him-sen-ammonia-dual-fuel-engine-with-maximum-output-of-5-4-mw-model-h32cdf-la
Japan Engine Corporation (J-ENG)	No known program	`7UEC50LSJA-HPSCR` -- 50 cm bore, 7 cyl, with high-pressure SCR after-treatment. Japan's first domestically developed large low-speed two-stroke to run on ammonia (co-firing operation Apr 2025). Official test runs late Aug 2025 at 100 % load / 95 % ammonia + 5 % pilot: N2O ~ 3 ppm; unburned NH3 virtually zero post-SCR; NOx ~ half of HFO mode; >90 % GHG reduction; thermal efficiency equal to or better than fuel-oil mode . First unit delivered to JMU Ariake Oct 2025 ; vessel (ammonia-fuelled medium gas carrier, NYK) in service Nov 2026. 60-bore version under development ; new ammonia-engine factory 2028 . Consortium: NYK, Nihon Shipyard, JMU, ClassNK.	No known program	https://ammoniaenergy.org/articles/j-eng-testing-complete-2-stroke-ammonia-engine-ready-to-roll-out/ https://www.j-eng.co.jp/en/news/20250421.html
Mitsui E&S	No known program	`7S60ME-C10.5-LGIA-HPSCR` -- Everllence-licensed ME-LGIA ammonia two-stroke with high-pressure SCR. Integrated land-based testing of engine + ammonia fuel-supply system completed Feb 2026 ; engine and fuel system moved to commercial stage . Expanding test and production facilities for mass production of both Everllence- and WinGD-licensed ammonia engines .	No known program	https://ammoniaenergy.org/articles/mitsui-es-commercializes-marine-ammonia-engine https://www.marinelink.com/news/mitsui-es-commercializes-ammonia-dual-542318
Rolls-Royce Power Systems (mtu)	No known program	No known program	No known ammonia engine program . mtu's alternative-fuel gas-engine roadmap is biogas/biomethane and hydrogen (up to 100 %) ; 2025-26 launches are fast-start natural-gas gensets for data centres (Series 4000 L64, 2.8 MW in 45 s, from 2026). *Ammonia not named as a product fuel.*	https://www.rolls-royce.com/media/press-releases/2025/02-10-2025-rr-introduces-fast-start-mtu-gas-gensets-for-powering-data-centers.aspx

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Bergen Engines (Langley Holdings; ex-Rolls-Royce)	No known program	No known program	`AMAZE` (Ammonia Zero Emission) research project -- launched Jan 2022, €4 M, ~3-year technology study led by Bergen with Equinor, SINTEF Energy, SINTEF Ocean, NTNU, RISE Fire Research; funded by Bergen + Research Council of Norway + Equinor. Goal: convert the Bergen diesel engine into a high-pressure multi-fuel unit with ammonia as primary fuel, seamlessly switchable to diesel/biofuel, ultimately as an economically viable retrofit package. *Status: research; no commercial ammonia product or date announced.*	https://www.bergenengines.com/about/sustainability/future-fuels/ammonia/ https://www.bergenengines.com/bergen-engines-launches-ambitious-ammonia-zero-emission-research-project/
Anglo Belgian Corporation (ABC) / BeHydro	No known program	No known program	No confirmed ammonia engine product. BeHydro (ABC + CMB.TECH JV) markets hydrogen dual-fuel and 100 % hydrogen engines, 900-2,670 kW, targeting shipping, rail and power generation. JV partner CMB.TECH works on hydrogen and ammonia applications, but no ammonia-fuelled ABC/BeHydro engine model was sourced -- treat as unconfirmed.	https://www.abc-engines.com/en/news/launch-behydro-si-hydrogen-engine
Cummins	No known program	No known program	No known ammonia engine program. Cummins' ammonia activity is upstream production (MoU with KBR on integrated green ammonia, Cummins supplying PEM electrolysis); its combustion R&D is hydrogen ICE (6.7 L / 15 L). *Ammonia not offered as an engine fuel.*	https://investor.cummins.com/news/detail/510/cummins-inc-signs-mou-with-kbr-to-collaborate-on
INNIO (Jenbacher)	No known program	No known program	No known ammonia program. Jenbacher gensets (250 kW-10.6 MW) are positioned as hydrogen-ready / up to 100 % H2; ammonia is not named as a fuel.	https://www.innio.com/en/news-media/press-releases/innio-jenbacher-gas-engines-ready-for-hydrogen/
Toyota Energy Solutions + AIST *(research pair, not a power-gen OEM)*	Pure-ammonia micro gas turbine, ~50 kW class -- kerosene-fired base machine re-combustored for bi-fuel kerosene/ammonia; world-first mono-fuel ammonia combustion in a gas turbine reported 2021. Follow-on development of a 300 kW ammonia combustion micro gas turbine. *Status: lab / demonstrator scale.*	No known program	No known program	https://ammoniaenergy.org/presentations/pure-ammonia-combustion-micro-gas-turbine-system/ https://link.springer.com/chapter/10.1007/978-981-19-4767-4_34

Cross-cutting read (evidence-based, no new numbers)

• **Gas turbines: two credible 100 %-NH3 tracks, both Japanese-anchored.**

IHI (2 MW IM270, ~3,000 h run, first commercial-plant deployment at PETRONAS/Gentari Terengganu from 2027) and **IHI + GE Vernova** (F-class, 2030). **MHI's H-25 40 MW** is the largest *nameplate* claim but the least recently evidenced. Everyone else (Siemens Energy, Ansaldo, Kawasaki, Doosan, Solar) is either hydrogen-first or paper-stage on ammonia.

• **Doosan and Siemens Energy define a third architecture:** don't burn NH3 -- **crack it and burn hydrogen**. That is directly relevant to MYSGH2's P1 (crack at MMHE, pipe H2) and should not be scored on the same axis as direct ammonia firing.

• ****Two-stroke engines are the most commercially advanced ammonia machines of any type**** -- Everllence ME-LGIA (type-approved Jun 2026, ~30+ engines ordered across the industry), WinGD X-DF-A (**in service** Jun 2026), J-ENG UEC50LSJA (test data published: N2O ~3 ppm, >90 % GHG cut). All ****marine propulsion****; none offered as a stationary genset.

• **Four-stroke is where stationary ammonia power actually shows up.** The only ammonia genset with an announced **commercial sales date** is ****IHI Power Systems' 6 MW-class 18V28ADF**** (sales from JFY2027, explicitly targeting data centres and industrial parks). **HD Hyundai** states an intent to serve the land power market with HiMSEN; **MAN/Everllence NH3 Spark** targets a **pilot-free retrofit** genset; **Bergen AMAZE** targets a retrofit package.

• **Malaysia is already on the map.** The **IHI-PETRONAS-Gentari 2 MW** ammonia-GT demonstration at Terengganu (start 2027) is the first ammonia-fired gas turbine deployment announced for a commercial plant **in-country** -- a direct, near-field precedent for MYSGH2 and a natural counterpart to Singapore's EMA/MPA 55-65 MW Jurong Island GT project.

Methodology & limitations

• **Sources.** OEM press releases and product pages first; then trade press (POWER Magazine, Turbomachinery Magazine, Gas Turbine World, Power Engineering, Riviera/Maritime Executive/Marine Log/Offshore Energy, LNG Prime, S&P Global) and the Ammonia Energy Association. Peer-reviewed sources used only where an OEM has published nothing (Ansaldo, Toyota/AIST).

• **Searches run 2026-09-10**; every status statement reflects the most recent item found on that date.

• **"No known program" means "no public program located in this survey"**, not "does not exist". Absence of evidence is reported as such -- see especially Kawasaki and Solar Turbines, both large, credible OEMs where a non-public/early program is entirely plausible.

• **Blocked domains.** The egress proxy blocked direct fetches of `ammoniaenergy.org`, `ihi.co.jp`, `wartsila.com`, `everllence.com` and

`siemens-energy.com`. Content from those domains was taken from search-engine extracts of the same pages, and the canonical URLs are still cited. **Where a load-bearing figure depends on one of these (notably the Wärtsilä 25 Ammonia delivery year), it is flagged as unresolved rather than asserted.**

• **Marine vs. stationary is not interchangeable.** Type approval for a marine genset is not a stationary power-plant product. The table states duty explicitly in every engine cell.

Open questions / needs verification

- [] **Wärtsilä 25 Ammonia first-delivery year: 2026 or 2028?** Two readings of the same 16 Apr 2026 release conflict. Needs the primary release re-read (domain currently blocked) or an OEM confirmation.
 - [] **MHI H-25 100 % NH3:** any 2024-2026 status update? Commercialisation date, and an **ammonia-specific** efficiency (the repo's 34.8 % anchor is still presumed NG-basis -- memory.md §3).
 - [] **IHI IM270 net electrical efficiency on 100 % ammonia** -- still not published by IHI in any source located.
 - [] **IHI Power Systems 18V28ADF:** net electrical efficiency, NOx/N2O/NH3-slip figures, after-treatment scope, and \$/kW. None published.
 - [] **HD Hyundai HiMSEN:** is there a *stationary* ammonia genset product (model, MW, date), or only the stated intent to enter that market?
 - [] **Hanwha Engine:** any ammonia two-stroke program of its own, distinct from the Hanwha Power Systems/Baker Hughes turbine work?
 - [] **Kawasaki and Solar Turbines:** confirm directly whether any ammonia turbine/engine program exists that is not publicly announced.
 - [] **Daihatsu (now DAIHATSU INFINEARTH):** trade summaries reference ammonia-fuelled four-stroke marine engines, but **no OEM source was located** -- deliberately excluded from the table rather than guessed.
 - [] **IHI-PETRONAS-Gentari Terengganu demo:** scope, offtake, ammonia source, and whether it creates a Malaysian precedent usable for MYSGH2 permitting.
 - [] **Everllence NH3 Spark:** target genset MW range and product date.
- *Prepared 2026-09-10 for the MYSGH2 `power\_generation/` workstream. Every cell is either sourced to the linked material, explicitly marked "no known program", or explicitly marked unconfirmed. No efficiency, cost, capacity or emissions figure has been inferred (CLAUDE.md §7).*